

ARCHIT MANGRULKAR

Bangalore, India

☎ 91- 8305414601 ✉ archit.mangrulkar@gmail.com [in LinkedIn](#) [GitHub](#) [CodeForces](#)

Education

Indian Institute of Technology Kharagpur

Nov 2020 – May 2024

Bachelor of Technology in Computer Science

CGPA 8.57/10.00

Work Experience

Samsung Research India - Bangalore

Jul 2024 – Present

C++ Developer | Visual Intelligence Team

Bangalore, India

- * Commercialized Video Stabilization, QuickTake, Hyperlapse & Astrolapse on Samsung's flagship S series smart phones
- * Mainlined camera pipelines for 60 fps video with **16.67 ms frame** window and support for 8K capture with 10-bit HDR
- * Leveraged ARM-specific SIMD, OpenCL optimizations & supported gyro sensor listener sampling at **526 microseconds**

University of Alberta, Canada

May 2023 – Aug 2023

Vision & Learning Lab | Multimodal Motion Capture

Edmonton, Canada

- * Developed large-scale multimodal dataset of humans in the wild of RGBD images, eye-tracking information, speech data
- * Programmed a **multithreaded C++ server** and Python GUI client using **PyQt5** to achieve concurrent image capture
- * Designed a 3D human pose tracking & clustering module for multi-view images with **YOLOv5**, DeepSORT, & DBSCAN

TU Dresden, Germany

May 2022 – Jul 2022

Chair of Processor Design | Approximate Computing in Neural Networks

Dresden, Germany

- * Designed neural operators as C static libraries for fast **8-bit FPGA** inference of neural networks using TensorFlow Lite
- * Explored cross-layer granularity combining signed 8-bit lookup tables, polynomial regression & EvoApproxLib multipliers
- * Reduced inference time by **2x** using **OpenMP's** multithreading, autovectorization, compiler flags & cache optimizations

Patents

Geometrically Consistent Multi-Camera Stabilization for XR Headsets (Patent Pending)

- Co-invented an AI driven multi-camera stabilization method enforcing geometric consistency for better visual experience
- Solved AR world-anchor loss during rapid, dynamic head motion through reinforcement learning inside stabilization loop
- Analyzed stabilization pipelines of **Apple**, **Meta**, and Xreal to pinpoint architectural limits & shape our novel claims

Key Projects

Unix Shell and Inter Process Communication

- Developed a robust Unix shell in C++ with process execution, I/O redirection, piping, & command history management
- Engineered **signal handlers** & implemented squashbug feature with custom heuristics for malware detection, termination
- Implemented shared memory graph with consumers using **3x** optimized **Dijkstra's** for shortest path calculation

MyHTTP TCP Browser

- Designed a concurrent CLI browser for file retrieval from **HTTP/HTTPS** servers with **OpenSSL** for secure sockets
- Developed an HTTP 1.1 server handling GET/PUT for PDFs, TXTs, JPGs with AccessLog and status code responses

SQL Query Rewriter Tool

- Developed a **Streamlit** web app for compiling and optimizing **relational algebra queries** with real-time result display
- Implemented a lexer and parser with **Lex & Yacc** to tokenize queries and define grammar rules for structured parsing
- Attained **3x** enhancement in processing time with a **C++** program featuring Node class for query tree representation

Publications

Real-World Video Representation Learning (VRL) | ECCV 26 (Under Submission)

- Deployed multi-branch video stability model in C++/ONNX runtime achieving a **15x speedup** over Python baseline
- Slashed **model size by 72%** via static INT8 quantization, solved Android Binder IPC limit & GPU memory contention
- Replaced RAFT model for optical flow with NeuFlow v2 demonstrating **44x FLOPS reduction** for edge deployment

Academic Achievements

- * Workshop paper on Stance & Premise Detection accepted at International Conference on **Computational Linguistics**
- * Offered department change to B.Tech CSE program for being among top 16 out of 1800 students securing 10.0 CGPA
- * **MITACS Scholarship** 2023 - University of Alberta, Canada | **Erasmus+ Scholarship** 2022 - TU Dresden, Germany
- * Secured **AIR 978** out of 150k candidates in JEE Advanced & **AIR 1517** out of 929k candidates in JEE Mains, 2020
- * Attained the silver medal in **DevRev's** Closed Domain QA Challenge at the **11th Inter IIT Tech Meet 2023**
- * Won the gold medal in **DRDO's** UAV Guided UGV Navigation Challenge at the **10th Inter IIT Tech Meet 2022**